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Lyapunov cohomology in random dynamical systems: theory and applications

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References

- LHD. Local stability for rough differential equations. (in preparation)
- LHD, P.T.Hong, N. D. Cong. Asymptotic stability for stochastic dissipative systems with a Hölder noise. (in preparation)
- N. D. Cong, LHD, P. T. Hong. Lyapunov spectrum of nonautonomous linear Young differential equations. arXiv:1807.02680
- LHD, M.J. Garrido-Atienza, A. Neuenkirch, B. Schmalfuss: Exponential stability of stochastic evolution equations driven by small fractional Brownian motion with Hurst parameter in $(1/2, 1)$, Journal of Differential Equations, Volume 264, 2018, 1119-1145.
- Nguyen Dinh Cong, LHD, Phan Thanh Hong: Nonautonomous Young differential equations revisited. J. Dynamics and Differential Equations, Volume 30 (4), 2018, 1921-1943.

Random dynamical systems

Definition

A (continuous) dynamical system on a metric space X is the triple $(\mathbb{R}, X, \pi)$ where $\pi : \mathbb{R} \times X \rightarrow X$, $(t, x) \mapsto \pi(t, x)$ is continuous and satisfies the following axioms:

$$i, \quad \pi(0, x_0) = x_0, \quad \forall x_0 \in X; \quad (1.1)$$

$$ii, \quad \pi(t + s, x_0) = \pi(t, \pi(s, x_0)), \quad \forall t, s \in \mathbb{R}, x_0 \in X. \quad (1.2)$$

Definition

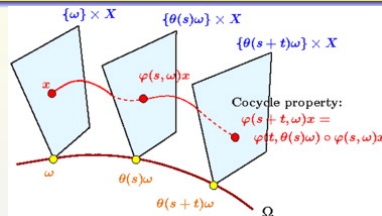
A (continuous) random dynamical system is an NDS (θ, φ) which in addition has the following properties:

(i), The driving system θ is an ergodic dynamical system $(\Omega, \mathcal{F}, \mathbb{P}, (\theta_t)_{t \in T})$, i.e., the base $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $(t, \omega) \mapsto \theta_t \omega$ is a measurable flow which is ergodic under $\mathbb{P}$.

ii, The cocycle $(t, \omega, x) \mapsto \varphi(t, \omega)x$ is measurable w.r.t $\mathcal{B} \otimes \mathcal{F} \otimes \mathcal{B}(X)$ and $\mathcal{B}(X)$, respectively, where $\mathcal{B}(X)$ is the Borel σ -algebra of X , such that the family $\varphi(t, \omega, \cdot) = \varphi(t, \omega) : X \rightarrow X$ of self-mappings of X satisfies the *cocycle property*

$$\varphi(0, \omega) = id_X, \quad \varphi(t + s, \omega) = \varphi(t, \theta_s \omega) \circ \varphi(s, \omega), \quad \forall t, s \in T, \quad \forall \omega \in \Omega. \quad (1.3)$$

Generation of RDS



- path-wise RDEs on $(\Omega, \mathcal{F}, \mathbb{P}, (\theta_t)_{t \in \mathbb{R}})$: G. Sell, V. Millionschikov, P. Kloeden, L. Arnold..

$$\dot{x} = f(\theta_t \omega, x). \quad (1.4)$$

- SDEs with standard Brownian motions: L. Arnold, P. Kloeden, T. Caraballo, B. Schmalfuss, H. Crauel, M. Scheutzow...

$$dX_t = f(X_t)dt + g(X_t)dW_t. \quad (1.5)$$

- SDEs with fractional Brownian motions: M. Hairer, B. Schmalfuss, M. Scheutzow,...(also with SPDE)

$$dX_t = f(X_t)dt + g(X_t)dB_t^H, H \in (0, 1). \quad (1.6)$$

Group valued cocycles

L. Arnold (1998): Random dynamical systems and cohomologies.

T. Tao (2008): Cohomology in dynamical systems.

An (abelian) cocycle is a function $\rho : G \times X \rightarrow U$ taking values in an abelian group $U = (U, +)$ that obeys the cocycle equation

$$\rho(gh, x) = \rho(h, x) + \rho(g, hx)$$

for all $g, h \in G, x \in X$.

Skew products flow (SPF) on $X \times_\rho U$ of the original dynamical system X , defined as the Cartesian product $\{(x, u) : x \in X, u \in U\}$ with the group action

$g(x, u) := (gx, u + \rho(g, x))$. Then $(gh)(x, u) = g(h(x, u))$.

A special type of cocycle is a coboundary

$$\rho : G \times X \rightarrow U, \quad \rho(g, x) := F(gx) - F(x)$$

for some function $F : X \rightarrow U$. A SPF $X \times_\rho U$ of a dynamical system by a coboundary $\rho(g, x) := F(gx) - F(x)$ can be conjugated to the trivial extension $X \times_0 U$ by the change of variables $(x, u) \mapsto (x, u - F(x))$.

Chains

For any $n \geq 0$, n -chain: formal integer linear combination of $n + 1$ -tuples $(g_1, \dots, g_n, x)$, where $x \in X$ and $g_1, \dots, g_n \in G$.

0-chain is a formal combination $\sum_{i=1}^m c_i x_i$ of points.

1-chain is a formal combination $\sum_{i=1}^m c_i (g_i, x_i)$ of line segments from x_i to $g_i x_i$, and so forth. Let $C_n(G, X)$ be the space of n -chains - an abelian group.

The boundary map

$$\partial : C_n(G, X) \rightarrow C_{n-1}(G, X)$$

to be the unique homomorphism such that

$$\begin{aligned} \partial(g_1, \dots, g_n, x) &= (g_1, \dots, g_{n-1}, g_n x) + \sum_{i=1}^{n-1} (-1)^{n-i} (g_1, \dots, g_{i-1}, g_i g_{i+1}, g_{i+2}, \dots, g_n, x) \\ &\quad + (-1)^n (g_2, \dots, g_n, x) \end{aligned}$$

thus $\partial^2 = 0$. For instance

$$\partial(g, x) = gx - x$$

$$\partial(g, h, x) = (g, hx) - (gh, x) + (h, x)$$

$$\partial(g, h, k, x) = (g, h, kx) - (g, hk, x) + (gh, k, x) - (h, k, x)$$

Cochains

An n -chain with vanishing boundary is called an n -cycle, while an n -chain which is the boundary of an $(n-1)$ -chain is called an n -boundary; the spaces of n -cycles and n -boundaries are denoted $Z_n(G, X)$ and $B_n(G, X)$ respectively.

We define the homology groups $H_n(G, X) := Z_n(G, X)/B_n(G, X)$ for all n .

An n -cochain is a homomorphism from the space $C_n(G, X)$ of n -chains to U . Since $C_n(G, X)$ is a free abelian group generated by the simplices $(x, g_1, \dots, g_n)$, we can view an n -cochain as a function $F : (x, g_1, \dots, g_n) \rightarrow F(x, g_1, \dots, g_n)$ from $G \times \dots \times G \times X \rightarrow U$. The space of all n -cochains is denoted

$C^n(G, X, U) := \text{Hom}(C_n(G, X), U)$ - an abelian group.

The boundary map $\partial : C_n(G, X) \rightarrow C_{n-1}(G, X)$ defines by duality a coboundary map $\delta : C^{n-1}(G, X, U) \rightarrow C^n(G, X, U)$, such that $\delta F(c) := F(\partial c)$ for all $F \in C^{n-1}(G, X, U)$ and $c \in C_n(G, X)$. Thus

$$\begin{aligned} \delta F(g_1, \dots, g_n, x) &= F(g_1, \dots, g_{n-1}, g_n x) + \sum_{i=1}^{n-1} (-1)^{n-i} F(g_1, \dots, g_{i-1}, g_i g_{i+1}, \dots, g_n, x) \\ &\quad + (-1)^n F(g_2, \dots, g_n, x). \end{aligned}$$

For instance: $\delta F(g, x) = F(gx) - F(x)$ for 0-cochains $F : X \rightarrow U$,

$\delta \rho(g, h, x) = \rho(g, hx) - \rho(gh, x) + \rho(h, x)$ for 1-cochains $\rho : G \times X \rightarrow U$.

Because $\partial^2 = 0$, we have $\delta^2 = 0$, and so $C^n(G, X, U)$ becomes a cochain complex. n -cochains whose coboundary vanishes are known as n -cocycles, and n -cochains which are the coboundary of an $(n - 1)$ -cochain are known as n -coboundaries. The spaces of n -cocycles and n -cochains are denoted $Z^n(G, X, U)$ and $B^n(G, X, U)$ respectively, allowing us to define the n^{th} cohomology group $H^n(G, X, U) := Z^n(G, X, U)/B^n(G, X, U)$.

When $n = 0$, the only 0-cocycles are the constants, and the only 0-coboundary is the zero function, so $H^0(G, X, U) \equiv U$.

When $n = 1$, the notion of 1-cocycle and 1-coboundary correspond to the notion of cycle and coboundary

Random attractors

Random sets: $\hat{M} = \{M(\omega)\}_{\omega \in \Omega}$ such that $\omega \mapsto d(x|M(\omega))$ is $\mathcal{F}$ -measurable for each $x \in X$.

Universe: $\mathcal{D}$ a family of random sets which is closed w.r.t. inclusions (i.e. if $\hat{D}_1 \in \mathcal{D}$ and $\hat{D}_2 \subset \hat{D}_1$ then $\hat{D}_2 \in \mathcal{D}$).

Definition (Crauel & Flandoli-1994, Schmalfuss-1998)

An invariant random compact set $\hat{A} \in \mathcal{D}$ is called

(i), a pullback random attractor in $\mathcal{D}$, if $\hat{A}$ attracts any closed random set $\hat{D} \in \mathcal{D}$ in the pullback sense, i.e.

$$\lim_{t \rightarrow \infty} d(\varphi(t, \theta_{-t}\omega)D(\theta_{-t}\omega)|A(\omega)) = 0; \quad (3.1)$$

(ii), a forward random attractor in $\mathcal{D}$, if $\hat{A}$ attracts any closed random set $\hat{D} \in \mathcal{D}$ in the forward sense, i.e.

$$\lim_{t \rightarrow \infty} d(\varphi(t, \omega)D(\omega)|A(\theta_t\omega)) = 0; \quad (3.2)$$

Pullback vs. Forward convergence

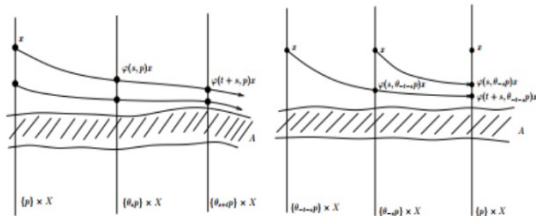


Figure 2.1: Forward attraction versus pullback attraction.

Example: $\dot{x} = \sin(t)x$: 0 is pullback attracting, but not forward attracting.

Existence of random attractor

Pullback absorbing set: $\hat{B} \in \mathcal{D}$ in a universe $\mathcal{D}$ such that $\hat{B}$ absorbs all sets in $\mathcal{D}$, i.e. for any $\hat{D} \in \mathcal{D}$, there exists a time $t_0 = t_0(\omega, \hat{D})$ such that

$$\varphi(t, \theta_{-t}\omega)D(\theta_{-t}\omega) \subset B(\omega), \text{ for all } t \geq t_0.$$

Theorem (Schenk-Hoppe 1998: Existence and Uniqueness of Random Attractor)

Given a universe $\mathcal{D}$, assume there exists a random compact pullback absorbing set $\hat{B} \in \mathcal{D}$. Then there exists a unique random pullback attractor (which is then a weak attractor) in $\mathcal{D}$, given by

$$A(\omega) = \bigcap_{t \geq 0} \varphi(t, \theta_{-t}\omega)B(\theta_{-t}\omega).$$

Application: prove that a system has an absorbing set.

Lyapunov exponents and Lyapunov spectrum

Definition

(i) The extended-real numbers (real numbers or symbol ∞ or $-\infty$)

$$\lambda_k(\omega) := \inf_{V \in \mathcal{G}_{d-k+1}} \sup_{y \in V} \limsup_{t \rightarrow \infty} \frac{1}{t} \log \|\Phi(t, \omega)y\|, \quad k = 1, \dots, d, \quad (3.3)$$

are called Lyapunov exponents of the flow $\Phi(t, \omega)$. The collection $\{\lambda_1(\omega), \dots, \lambda_d(\omega)\}$ is called Lyapunov spectrum of the flow $\Phi(t, \omega)$.

(ii) The linear subspaces of $\mathbb{R}^d$

$$E_k(\omega) := \{y \in \mathbb{R}^d \mid \limsup_{t \rightarrow \infty} \frac{1}{t} \log \|\Phi(t, \omega)y\| \leq \lambda_k(\omega)\}, \quad k = 1, \dots, d, \quad \omega \in \Omega, \quad (3.4)$$

are called Lyapunov subspaces at time u of the flow $\Phi(t, \omega)$. The flag of nonincreasing linear subspaces of $\mathbb{R}^d$ $\mathbb{R}^d = E_1^u(\omega) \supset E_2^u(\omega) \supset \dots \supset E_d^u(\omega) \supset \{0\}$ is called Lyapunov flag at time u of the flow $\Phi(t, \omega)$.

Theorem

- i, Let $\Phi(t, \omega)$ be a linear RDS on $\mathbb{R}^d$ on the time interval $[0, \infty)$. Then the Lyapunov exponents $\lambda_k(\omega)$, $k = 1, \dots, d$, of $\Phi(t, \omega)$ are measurable functions of $\omega \in \Omega$.
- ii, The Lyapunov subspaces $E_k(\omega)$, $k = 1, \dots, d$, of $\Phi(t, \omega)$ are measurable with respect to $\omega \in \Omega$, and invariant with respect to the flow in the following sense

$$\Phi(t, \omega)E_k(\omega) = E_k(\theta_t \omega), \quad \forall t \in [0, \infty), \omega \in \Omega, k = 1, \dots, d.$$

Theorem

Let $\Phi(t, \omega)$ be the RDS generated by the SDE driven by fBm $\{\lambda_1(\omega), \dots, \lambda_d(\omega)\}$ be the Lyapunov spectrum of the flow $\Phi(t, \omega)$. Then the Lyapunov exponents $\lambda_k(\omega)$, $k = 1, \dots, d$, can be computed via a discrete-time interpolation of the flow, i.e.

$$\lambda_k(\omega) := \inf_{V \in \mathcal{G}_{d-k+1}} \sup_{y \in V} \limsup_{N \ni t \rightarrow \infty} \frac{1}{t} \log \|\Phi(t, \omega)y\|, \quad k = 1, \dots, d. \quad (3.5)$$

Moreover, the random variables $\lambda_1(\omega), \dots, \lambda_d(\omega)$ have finite moments of order p for any $p \geq 1$. Hence, in particular, they are finite almost surely.

Cohomologous cocycles

- Skew product flow $\Theta : T \times \Omega \times X \rightarrow \Omega \times X$ such that $\Theta(t, \omega, x) = (\theta_t \omega, \varphi(t, \omega)x)$.
- Measurable bundle $\psi(\omega) := \Psi|_{\pi^{-1}(\omega)} : \pi^{-1}(\omega) \rightarrow \{\omega\} \times X$
- Bundle RDS: $\pi \circ \Theta_t = \theta_t \circ \pi$, or $\varphi(t, \omega) : \Theta_t|_{\pi^{-1}(\omega)} : \pi^{-1}(\omega) \rightarrow \pi^{-1}(\theta_t \omega)$ forms a cocycle.
- Isomorphism of bundle RDS: there exists a measurable isomorphism Ψ such that $\Theta_t^2 \circ \Psi = \Psi \circ \Theta_t^1$
- Cohomology of cocycles: $\Theta_t^2 \circ \Psi = \Psi \circ \Theta_t^1$ iff $\varphi_2(t, \omega) \circ \psi(\omega) = \psi(\theta_t \omega) \circ \varphi_1(t, \omega)$.
- Lyapunov cohomology: $\psi(\omega) : (\pi_1^{-1}(\omega), \|\cdot\|_\omega^1) \rightarrow (\pi_2^{-1}(\omega), \|\cdot\|_\omega^2)$ is cohomology and satisfies the temperedness

$$\lim_{t \rightarrow \pm\infty} \frac{1}{t} \log \|\psi(\theta_t \omega)\|_{\theta_t \omega} = \lim_{t \rightarrow \pm\infty} \frac{1}{t} \log \|\psi(\theta_t \omega)^{-1}\|_{\theta_t \omega} = 0.$$

- Lyapunov spectrum is perserved under Lyapunov cohomology, i.e. $\mathcal{S}(\theta, \Phi_2) = \mathcal{S}(\theta, \Phi_1)$, where $\psi(\omega)E_i^1(\omega) = E_i^2(\omega)$.
- Random attractors are perserved under Lyapunov cohomology, i.e. $\psi(\omega)\mathcal{A}_1(\omega) = \mathcal{A}_2(\omega)$.

Conjugacy transformation: Doss-Sussmann technique

- Sussmann (1978): there exists an explicit transformation which transforms

$$dX_t = F(X_t)dt + G(X_t)dB(t) \text{ into} \quad (3.6)$$

$$dY_t = f(Y_t) + \sigma dB(t) \quad (3.7)$$

- Imkeller-Schmalfuss (1998), Imkeller-Lederer(2001): there exists an explicit transformation which transforms (3.6) into a random differential equation

$$\frac{dY_t}{dt} = f(\theta_t\omega, Y_t) \quad (3.8)$$

- Mohammed-Scheutzov (2003): there exists an explicit transformationn which transform

$$dX_t = F(X_t)dt + G(X_t)dB(t) \text{ into (3.8).} \quad (3.9)$$

- Question: Does there exist an explicit transformation for

$$dX_t = F(X_t)dt + G(X_t)dB^H(t) \text{ into (3.8) or}$$

$$dY_t = f(Y_t) + \sigma dB^H(t)?$$

Given $H \in (0, 1)$, a continuous centered Gaussian process $B^H = (B^H(t), t \in \mathbb{R})$, with covariance function

$$R(s, t) = \frac{1}{2}(|t|^{2H} + |s|^{2H} - |t - s|^{2H}), \quad s, t \in \mathbb{R},$$

on an appropriate probability space, is called a two-sided one-dimensional fractional Brownian motion (fBm) with Hurst parameter H . When $H = 1/2$, B is the standard Brownian motion.

We consider a canonical version of this process given by the probability space $(C_0(\mathbb{R}; \mathbb{R}^d), \mathcal{B}(C_0(\mathbb{R}; \mathbb{R}^d)), \mathbb{P})$ where $\mathbb{P}$ is the Gauß-measure generated by W . On this probability space we can also introduce the shift

$$\theta_t \omega(\cdot) = \omega(\cdot + t) - \omega(t).$$

It is known that $(C_0(\mathbb{R}; \mathbb{R}^d), \mathcal{B}(C_0(\mathbb{R}; \mathbb{R}^d)), \mathbb{P}, \theta)$ is an ergodic metric dynamical system. fractional Brownian motion B^H using the representation in Mandelbrot and van Ness

$$B^H(t) := \frac{1}{c_H} \left(\int_{-\infty}^0 [(t-u)^{H-\frac{1}{2}} - (-u)^{H-\frac{1}{2}}] dW(u) + \int_0^t (t-u)^{H-\frac{1}{2}} dW(u) \right),$$

with

$$c_H := \left(\frac{1}{2H} + \int_0^\infty [(1+u)^{H-\frac{1}{2}} - u^{H-\frac{1}{2}}]^2 du \right)^{\frac{1}{2}}.$$

p-var norms, Hölder norms

Let $0 \leq T_1 < T_2 < \infty$, $C^{r\text{-var}}([T_1, T_2], \mathbb{R}^d)$ be the Banach space of bounded r -variation continuous functions on $[T_1, T_2]$ having values in $\mathbb{R}^d$ with finite norm

$$\|u\|_{r\text{-var}, [T_1, T_2]} = |u_{T_1}| + \|u\|_{r\text{-var}, [T_1, T_2]} < \infty,$$

where

$$\|u\|_{p\text{-var}, [T_1, T_2]} = \left(\sup_{\Pi(T_1, T_2)} \sum_{i=0}^{n-1} |u_{t_{i+1}} - u_{t_i}|^r \right)^{1/r}$$

For each $0 < \alpha < 1$, we denote by $C^{\alpha\text{-Hol}}([T_1, T_2], \mathbb{R}^d)$ the space of Hölder continuous functions on $[T_1, T_2]$ equipped with the norm

$$\|u\|_{\alpha\text{-Hol}, [T_1, T_2]} := |u_{T_1}| + \sup_{T_1 \leq s < t \leq T_2} \frac{|u(t) - u(s)|}{(t - s)^\alpha}.$$

We also denote by $\|\cdot\|_{\infty, [T_1, T_2]}$ the supnorm of continuous functions on $[T_1, T_2]$. It is obvious that $\forall u \in C^{\alpha\text{-Hol}}([T_1, T_2], \mathbb{R}^d)$ with $\alpha = 1/r$.

$$\|u\|_{r\text{-var}, [T_1, T_2]} \leq (T_2 - T_1)^\alpha \|u\|_{\alpha\text{-Hol}, [T_1, T_2]}$$

$H > \frac{1}{2}$: Young integrals

Now, consider $x \in C^{q-\text{var}}([T_1, T_2], \mathbb{R}^{d \times m})$ and $\omega \in C^{p-\text{var}}([T_1, T_2], \mathbb{R}^m)$, $p, q \geq 1$, if Riemann-Stieltjes sums for finite partition $\Pi = \{T_1 = t_0 < t_1 < \dots < t_n = T_2\}$ of $[T_1, T_2]$ and any $\xi_i \in [t_i, t_{i+1}]$

$$S_\Pi := \sum_{i=0}^{n-1} x(\xi_i)(\omega(t_{i+1}) - \omega(t_i)),$$

converges as the mesh $|\Pi| := \max_{0 \leq i \leq n-1} |t_{i+1} - t_i|$ tends to zero, we call the limit is the

Young integral of x w.r.t ω on $[T_1, T_2]$ denoted by $\int_{T_1}^{T_2} x(t) d\omega(t)$. It is well known that if $p, q \geq 1$ and $\frac{1}{p} + \frac{1}{q} > 1$, the Young integral $\int_a^b x_t d\omega_t$ exists, and satisfies the so-called Young-Loeve estimate

$$\left| \int_s^t x(u) d\omega(u) - x(s)[\omega(t) - \omega(s)] \right| \leq K \|x\|_{q-\text{var}, [s, t]} \|\omega\|_{p-\text{var}, [s, t]}, \quad (3.10)$$

where

$$K := (1 - 2^{1-\theta})^{-1}, \quad \theta := \frac{1}{p} + \frac{1}{q} > 1. \quad (3.11)$$

$\frac{1}{3} < H < \frac{1}{2}$: rough path integrals

We also introduce the construction of the integral using rough paths for the case $x, \omega \in C^\alpha([a, b])$ when $\alpha \in (\frac{1}{3}, \frac{1}{2})$. To do that, we need to introduce the concept of rough paths. Namely, a couple $(\omega, \mathbb{W})$ with $\omega \in C^\alpha([a, b])$, $\mathbb{W} \in C^{2\alpha}(\Delta^2[a, b])$, $\Delta^2[a, b] := \{(s, t) : a \leq s \leq t \leq b\}$, is called a rough path if they satisfies Chen's relation

$$\mathbb{W}_{s,t} - \mathbb{W}_{s,u} - \mathbb{W}_{u,t} = (\omega_t - \omega_u) \otimes (\omega_u - \omega_s), \quad \forall a \leq s \leq u \leq t \leq b.$$

Denote by $C^\alpha([a, b])$ the space of all rough paths in $[a, b]$. A path $x \in C^\alpha([a, b])$ is then called to be *controlled by* ω if there exists a tube (x', R^x) with $x' \in C^\alpha([a, b])$, $R^x \in C^{2\alpha}(\Delta^2[a, b])$ such that

$$x_t - x_s = x'(\omega_t - \omega_s) + R^x_{s,t}, \quad \forall a \leq s \leq t \leq b.$$

The space $\mathcal{D}_\omega^{2\alpha}([a, b], \mathbb{R}^d)$ of all the couple (x, x') that is controlled by ω will be a Banach space equipped with the norm

$$\begin{aligned} \|x, x'\|_{\omega, 2\alpha} &:= \|x_a\| + \|x'_a\| + |||x, x' |||_{\omega, 2\alpha}, & \text{where} \\ |||x, x' |||_{\omega, 2\alpha} &:= |||x' |||_{\alpha, [a, b]} + |||R^x |||_{2\alpha, [a, b]}. \end{aligned}$$

$\frac{1}{3} < H < \frac{1}{2}$: rough path integrals

Now fix a rough path $(\omega, \mathbb{W})$, then for any $(x, x') \in \mathcal{D}_\omega^{2\alpha}[a, b]$, it can be proved that the function $F \in C^\alpha(\Delta^2[a, b])$ defined by

$$F_{s,t} := x_s(\omega_t - \omega_s) + x'_s \mathbb{W}_{s,t}$$

belongs to the space

$$C_2^{\alpha, 3\alpha}([a, b]) := \left\{ F \in C^\alpha(\Delta^2[a, b]) : F_{t,t} = 0 \text{ and } \|\delta F\|_{3\alpha, [a, b]} := \sup_{a \leq s \leq u \leq t \leq b} \frac{\|F_{s,t} - F_{s,u} - F_{u,t}\|}{|t - s|^{3\alpha}} < \infty \right\}.$$

Thanks to the sewing lemma, the integral $\int_s^t x_u d\omega_u$ can be defined as

$$\int_s^t x_u d\omega_u := \lim_{|\Pi| \rightarrow 0} \sum_{[u, v] \in \Pi} [x_u(\omega_v - \omega_u) + x'_u \mathbb{W}_{u, v}]$$

where the limit is taken on all the finite partition Π of $[a, b]$ with $|\Pi| := \max_{[u, v] \in \Pi} |v - u|$.

Moreover, there exists a constant $C_\alpha = C(\alpha, b - a) > 1$ such that

$$\left\| \int_s^t x_u d\omega_u - x_s(\omega_t - \omega_s) - x'_s \mathbb{W}_{s, t} \right\| \leq C_\alpha |t - s|^{3\alpha} \left(\|\omega\|_{\alpha, [s, t]} \|R^x\|_{2\alpha, \Delta^2[s, t]} + \|x'\|_{\alpha, [a, b]} \|\mathbb{W}\|_{2\alpha, [a, b]} \right) \quad (3.12)$$

$\frac{1}{3} < H < \frac{1}{2}$: rough path integrals

From now on, if no other emphasis, we will simply write $\|x\|_\alpha$ or $\|\mathbb{W}\|_{2\alpha}$ without addressing the domain in $[a, b]$ or $\Delta^2[a, b]$. In particular, for any $f \in C_b^2(\mathbb{R}^d, \mathbb{R}^d)$, then $f(\omega) \in \mathcal{D}_\omega^{2\alpha}$ with $f(\omega)' = \nabla f(\omega)$ and

$$\|f(\omega), \nabla f(\omega)\|_{\omega, 2\alpha} \leq \|\nabla^2\|_\infty \left(\|\omega\|_\alpha + \frac{1}{2} \|\omega\|_\alpha^2 \right).$$

In that case (3.12) becomes

$$\left| \int_s^t f(\omega_u) d\omega_u - f(\omega_s)(\omega_t - \omega_s) + \nabla f(\omega_s) \mathbb{W}_{s,t} \right| \leq C |t-s|^{3\alpha} \|f\|_{C_b^2} \left(\|\omega\|_\alpha^3 + \|\omega\|_\alpha \|\mathbb{W}\|_{2\alpha} \right).$$

Moreover, in case $f \in C_b^3$ then we get the formula for integration by composition

$$f(\omega_t) = f(\omega_s) + \int_s^t \nabla f(\omega_u) d\omega_u + \frac{1}{2} \int_s^t \nabla^2 f(\omega_u) d[\omega]_{s,u},$$

where the last integral is understood in the Young sense and

$$[\omega]_{s,t} := (\omega_t - \omega_s) \otimes (\omega_t - \omega_s) - 2 \operatorname{Sym} (\mathbb{W}_{s,t}) \in C^{2\alpha}.$$

Theorem

Under the regularity conditions of coefficient functions, there exists a transformation which transform

$$dX_t = F(X_t)dt + G(X_t)dB^H(t) \quad (3.13)$$

into a random differential delay equation which is conjugated to (3.13).

Sketch of the proof: $z(t, 0, z_0)$ - the solution of the autonomous differential equation $\dot{z} = G(z)$ generates a C^1 semigroup $\varphi(\cdot, \cdot) : \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ defined by $\varphi(t, z_0) = z(t, 0, z_0)$. It also follows from the autonomy that

$$\frac{d\varphi}{dz}(t, \varphi(t, z_0))G(z_0) = G(\varphi(t, z_0)). \quad (3.14)$$

It can be proved that

$$\varphi(\omega(t), z_0) - z_0 = \int_0^t \frac{d\varphi}{ds}(\omega(s), z_0) d\omega(s) = \int_0^t G(\varphi(\omega(s), z_0)) d\omega(s).$$

Hence, $\varphi(\omega(t), z_0)$ is the solution of the fractional SDE $dy(t) = G(y(t))d\omega(t)$. We introduce the following transformation $X(t) = \varphi(\omega(t), u(t))$. Then we can prove that

$$u(t) - u(0) = \int_0^t \frac{d\varphi}{dx}(-\omega(s), X_s) F(X_s) ds,$$

where the last equality is due to (3.14). This proves that $u(\cdot)$ is differentiable w.r.t. t and satisfies the following random differential delay equation

$$\dot{u}(t) = \frac{d\varphi}{dz}(-\omega(t), \varphi(\omega(t), u(t))) f(\varphi(\omega(t), u(t))). \quad (3.15)$$

Consider for parameter $\mu > 0$ the fractional Langevin equation

$$dz(t) = -\mu z(t)dt + dB^H(t). \quad (3.16)$$

Lemma

Equation (3.16) possesses a solution which can be computed explicitly as

$$z_\mu(t, \omega) = B^H(t, \omega) - \mu e^{-\mu t} \int_{-\infty}^t e^{\mu u} B^H(u, \omega) du, \quad t \in \mathbb{R}. \quad (3.17)$$

The solution z_μ is stationary, i.e. $z_\mu(t, \omega) = z_\mu(0, \theta_t \omega)$. Moreover, $E z_\mu(0, \cdot)^2 < C \mu^{-2H}$ for some constant $C > 0$ independent of μ , in particular there exists a subsequence of parameters $\mu_n \rightarrow \infty$ such that $z_{\mu_n}(0, \omega) \rightarrow 0$ as $n \rightarrow \infty$ a.s.

Sketch of the proof: First, denote $D(\mathbb{R}^d)$ the Polish space of diffeomorphisms of $\mathbb{R}^d$. Then $\psi(0, \omega) := \varphi(z_\mu(0, \omega), \cdot) \in D(\mathbb{R}^d)$ and $\psi(t, \omega) := \varphi(z_\mu(t, \omega), \cdot) \in D(\mathbb{R}^d)$ satisfies the $D(\mathbb{R}^d)$ -valued SDE $d\psi(t) = G(\psi(t))dz_\mu(t)$.

Next, introduce the transformation $x(t, \omega) = \psi(t, \omega)y(t, \omega)$.

Random attractors for dissipative systems

We would like to investigate the SDE of the form

$$d\mathbf{X}_t = f(\mathbf{X}_t)dt + h(\mathbf{X}_t)dB_t^H, \quad (4.1)$$

where $h(\mathbf{X}) = \sigma\mathbf{X}$ for the multiplicative noise and when $H = \frac{1}{2}$, (4.1) is Stratonovich SDE with respect to the Brownian motion B_t . The drift function f satisfies the general dissipativity condition

$$\langle \mathbf{X} - \mathbf{Y}, f(\mathbf{X}) - f(\mathbf{Y}) \rangle \leq a - b\|\mathbf{X} - \mathbf{Y}\|^2, \quad a, b > 0. \quad (4.2)$$

We are going to prove that

Theorem

There exists a random attractor for (4.1).

$H = \frac{1}{2}$: Arnold & Schmalfuss (1996), Crauel & Flandoli (1994)

$H \neq \frac{1}{2}$: Garrido-Atienza & Neuenkirch & Kloeden (2009), Gess et al. (2011) for SPDE with additive noise.

Idea: transform the SDE into a conjugated RDE and then prove that there exists an absorbing set.

Sketch of the proof

Multiplicative noise: introduce the transformation $\mathbf{X}_t = e^{\sigma Z_t} \mathbf{Y}_t$ where z_t is the unique stationary solution of the Langevin equation

$$dz_t = -z_t dt + dB_t.$$

By the integration by part rule, it follows that

$$d\mathbf{X}_t = \sigma \mathbf{X}_t dz_t + e^{\sigma Z_t} \mathbf{Y}_t,$$

therefore system (4.1) will be transformed into a random differential equation

$$\dot{\mathbf{Y}}_t = g(z_t, \mathbf{Y}_t) := e^{-\sigma Z_t} f(e^{\sigma Z_t} \mathbf{Y}_t) + \sigma z_t \mathbf{Y}_t. \quad (4.3)$$

Observe that

$$\begin{aligned} \frac{d}{dt} \|\mathbf{Y}_t\|^2 &= 2 \langle \mathbf{Y}_t, g(z_t, \mathbf{Y}_t) \rangle \\ &= 2e^{-2\sigma Z_t} \langle e^{\sigma Z_t} \mathbf{Y}_t, f(e^{\sigma Z_t} \mathbf{Y}_t) + 2\sigma z_t \|\mathbf{Y}_t\|^2 \rangle \\ &\leq 2e^{-2\sigma Z_t} [a - b \|e^{\sigma Z_t} \mathbf{Y}_t\|^2 + \langle f(0), e^{\sigma Z_t} \mathbf{Y}_t \rangle] + 2\sigma z_t \|\mathbf{Y}_t\|^2 \\ &\leq \left[2a + \frac{\|f(0)\|^2}{b} \right] e^{-2\sigma Z_t} + (2\sigma z_t - b) \|\mathbf{Y}_t\|^2 =: p(z_t) + q(z_t) \|\mathbf{Y}_t\|^2. \end{aligned}$$

Hence by the comparison principle, $\|\mathbf{Y}_t\|^2 \leq R_t$ whenever $\|\mathbf{Y}_0\|^2 \leq R_0$ where R_t is the solution of

$$\dot{R}_t = p(z_t) + q(z_t)R_t, \quad (4.4)$$

which can be computed explicitly as

$$R_t(\omega, R_0) = e^{\int_0^t q(z_u(\omega)) du} R_0 + \int_0^t p(z_s(\omega)) e^{\int_s^t q(z_u(\omega)) du} ds.$$

Observe that by the Birkhorff ergodic theorem, there exist almost surely

$$\lim_{t \rightarrow -\infty} \frac{1}{t} \int_t^0 q(z_u) du = \lim_{t \rightarrow -\infty} \frac{1}{t} \int_t^0 q(z(\theta_u \omega)) = E q(z(\cdot)) = -b < 0,$$

hence there exists a unique stationary solution of (4.4) which can be written in the form

$$\bar{R}(\omega) = \int_{-\infty}^0 p(z_s(\omega)) e^{\int_s^0 q(z_u(\omega)) du} ds.$$

Moreover $\|\mathbf{Y}_t(\omega, \mathbf{Y}_0)\|^2 \leq \bar{R}(\theta_t \omega)$ whenever $\|\mathbf{Y}_0\|^2 \leq \bar{R}(\omega)$ and

$$\limsup_{t \rightarrow \infty} \|\mathbf{Y}_t(\theta_{-t} \omega, \mathbf{Y}_0)\|^2 \leq \limsup_{t \rightarrow \infty} R_t(\theta_{-t} \omega, R_0) = \bar{R}(\omega).$$

Hence the ball $B(0, R(\omega))$ is actually forward invariant under the random dynamical system generated by (4.3) and is also a pullback absorbing set. Thus there exists a random attractor for (4.3). Finally, since systems (4.1) and (4.3) are conjugate under the random linear transformation $\mathbf{X}_t = e^{\sigma Z_t} \mathbf{Y}_t$ which is tempered, there exists also a random attractor for system (4.1).

Random attractor for strictly dissipative systems

We would like to investigate the SDE of the form

$$d\mathbf{X}_t = f(\mathbf{X}_t)dt + g(\mathbf{X}_t)dB_t^H, \mathbf{X} \in \mathbb{R}^d \quad (4.5)$$

where f satisfies the strict dissipativity condition

$$\langle \mathbf{X} - \mathbf{Y}, f(\mathbf{X}) - f(\mathbf{Y}) \rangle \leq -b\|\mathbf{X} - \mathbf{Y}\|^2, b > 0, \forall \mathbf{X}, \mathbf{Y} \in \mathbb{R}^d \quad (4.6)$$

and g satisfies the global Lipschitz continuity.

$$\|g(\mathbf{X}) - g(\mathbf{Y})\| \leq c_g\|\mathbf{X} - \mathbf{Y}\|, \forall \mathbf{X}, \mathbf{Y} \in \mathbb{R}^d. \quad (4.7)$$

For the simple case, assume that $f(0) = g(0) = 0$.

- $H = \frac{1}{2}$: Mao (2009), Kloeden & Wu (2011) proves for SDE and SDDE, $\mathbf{X}_t$ converge to 0 exponentially in the mean square sense, thus also in the pathwise sense.
- $H > \frac{1}{2}$: LHD & Garrido-Atienza & Neuenkirch & Schmalfuss (2018) proves for SPDE, criteria quite complicated. For $\mathbb{R}^d$ it is reduced to system

$$d\mathbf{X}_t = [A\mathbf{X}_t + F(\mathbf{X}_t)]dt + G(\mathbf{X}_t)dB_t^H, \mathbf{X} \in \mathbb{R}^d \quad (4.8)$$

where A -negative definite with $-\lambda$, and F is globally Lipschitz continuous with c_{DF} . $G \in C^1$, globally Lipschitz continuous with c_{DG} .

Definition

(A) **Stability:** A solution $\mu(\cdot)$ of a differential equation is called stable, if for any $\varepsilon > 0$ there exists an $r = r(\varepsilon) > 0$ such that for any solution $x(\cdot)$ satisfying $\|x(0) - \mu(0)\| < r$ the following inequality holds

$$\sup_{t \geq 0} \|x(t) - \mu(t)\| < \varepsilon.$$

(B) **Attractivity:** $\mu(\cdot)$ is called attractive, if there exists $r > 0$ such that for any solution $x(\cdot)$ satisfying $\|x(0) - \mu(0)\| < r$ we have

$$\lim_{t \rightarrow \infty} \|x(t) - \mu(t)\| = 0.$$

(C) **Asymptotic stability:** $\mu(\cdot)$ is called

- ① asymptotically stable, if it is stable and attractive.
- ② exponentially stable, if it is stable and there exists $r > 0$ such that for any solution $x(\cdot)$ satisfying $\|x(0) - \mu(0)\| < r$ we have

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \|x(t) - \mu(t)\| < 0.$$

Consider the following rough differential equation

$$dx_t = [Ax_t + f(x_t)]dt + [Cx_t + g(x_t)]d\omega_t, \quad (4.9)$$

or in the integral form

$$x_t = x_a + \int_a^t [Ax_u + f(x_u)]du + \int_a^t [Cx_u + g(x_u)]d\omega_u, \quad t \in [a, T]; \quad (4.10)$$

To study the local stability, we impose conditions for matrices $A, C \in \mathbb{R}^{d \times d}$ such that A is negative definite, i.e. there exists a $\lambda > 0$ such that

$$\langle x, Ax \rangle \leq -\lambda \|x\|^2. \quad (4.11)$$

The nonlinear part $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is locally Lipschitz function such that

$$f(0) = 0 \quad \text{and} \quad \|f(x)\| \leq \|x\|k(\|x\|) \quad (4.12)$$

where $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is an increasing function. In this paper, we also assume that g has a special form

$$g(0) = 0 \quad \text{and} \quad g(x) = \|x\| G\left(\frac{x}{\|x\|}\right), \quad \forall x \neq 0, \quad (4.13)$$

where $G \in C^1$ has a bound in the unit circle, i.e. $\sup_{\|y\|=1} \|G(y)\| = C_G^\infty < \infty$, and also has bounded derivatives C_{DG} (which also include the Lipschitz coefficient of the highest derivative).

Assign

$$C_G := \max\{C_G^\infty, C_{DG}\}.$$

Theorem (LHD - 2018)

Assume conditions (4.11), (4.12) are satisfied, and moreover $\lambda > k(0)$. Then there exists an $\epsilon > 0$ such that given $\tilde{C} := \|C\| + C_G < \epsilon$, the zero solution of (4.9) is locally exponentially stable.

Our scheme is to estimate the norm growth of the solution in discrete intervals using the rough estimates of the angular rough equation. Using the comparison principle, it is then sufficient to study the local stability of the corresponding random differential equation, which can be done with random norm techniques

Thank you for your attention!